

Wojciech Trejter

github.com/Voytech-0
linkedin.com/in/wojciech-trejter
Rotterdam, Netherlands

Summary

AI engineer who turns research and data into working software: Python machine learning models, simulation pipelines, and full-stack tools, applied across computer vision, robotics, and interpretability. MSc Artificial Intelligence (Cum Laude, University of Amsterdam), author in *Transactions on Machine Learning Research*, winner of the European Rover Challenge 2023, two-time ICPC competitor.

Work History

AI Intern **xBerry** *May – Aug 2026, Wroclaw*

- Built a sim-to-real transfer pipeline between Isaac Sim/Isaac Lab simulation and real robotic hardware; the team raised real-robot task success from 0% to 80%.
- Fine-tuned world models with LoRA (Low-Rank Adaptation), adapting large models within a limited compute budget.
- Architected simulation pipelines for training and evaluating high-fidelity robotic arm tasks in Isaac Lab.

Teaching Assistant **University of Amsterdam** *Sep 2025 – Apr 2026, Amsterdam*

- Led seminars and tutorials for the Computer Vision and Reinforcement Learning courses, turning abstract concepts (convolutional filters, policy gradients) into example-driven explanations.
- Graded coursework and mentored students on assignments, targeting feedback at commonly misunderstood theory.

Software Engineer Intern **IBM CIC** *Feb 2024 – Jun 2024, Groningen*

- Delivered an internal full-stack dashboard (MongoDB, Express, React, Node — MERN) end-to-end, from database schema to UI.
- Worked in a Scrum team on bi-weekly sprints, running standups and sprint reviews.

Technical Team Lead **Makercie** *Jan 2023 – Jun 2024, Groningen*

- Took **1st place** at the European Rover Challenge Remote Edition (2023), leading a 7-person software team (Jan – Sep 2023); then team member until Jun 2024.
- Built the rover's autonomy stack in Robot Operating System (ROS): computer-vision-driven control of a 7-DoF robotic arm, tested in Gazebo simulation before deployment on the rover.

Education

MSc Artificial Intelligence **University of Amsterdam** *Sep 2024 – Aug 2026, Amsterdam*

- **Cum Laude** with GPA 8.3/10.
- **Thesis**: *Evo-SLAM: Monocular Gaussian Splatting SLAM for Evolving Scenes*, an online system built on Vision Transformers and foundation models, outperforming published state-of-the-art methods (GaME, WildGS-SLAM); paper under review.
- Courses: Computer Vision 1&2, Deep Learning 1&2, Machine Learning 1&2, NLP, MLOps.

BSc Artificial Intelligence **University of Groningen** *Sep 2021 – Jul 2024, Groningen*

- **Cum Laude** with GPA 8.8/10 and 270 ECTS (180 core + 45 Honours + 45 extracurricular).
- Finished with Honours; Honours project on deep-learning image classification for Portable Antiquities Netherlands (see Projects).
- **Thesis**: *Application of Multitask Learning Paradigm on Hydrodynamic Imaging*.
- Courses: Probability Theory, Neural Networks, Computer Graphics, Computational Complexity.

Publications

- Kurek, I., **Trejter, W.**, Frkovic, S., Erdelez, A., [Re] Improving Interpretation Faithfulness for Vision Transformers, *Transactions on Machine Learning Research 2025*.
  - Reproduced a published method for faithful Vision Transformer explanations; the results did not support the original paper's claims.
  - Ran experiments on an HPC cluster (SLURM).
- **Trejter, W.**, Wolf, B. J., Application of Multitask Learning Paradigm on Hydrodynamic Imaging, BSc thesis.
  - Multitask deep learning (MLP, LSTM) on flow-sensor data for object localisation and shape classification.

Projects

- **SAR Ship Locator** (2026): Weekend project to learn SAR basics: Sentinel-1 ship detection checked against AIS.
- **GAPC** (Groningen Algorithmic Programming Contest, 2023): Organised the contest and designed its problem set.
- **Image Classification for Antiquities**: Honours project: image classification by fine-tuning Residual Networks (ResNets) on a dataset obtained from Portable Antiquities Netherlands.
- **Rodina AI Startup**: Co-founder and programmer for a B2B startup creating synthetic image datasets; built and delivered an MVP to a client.

Awards

- **European Rover Challenge Remote Edition (2023)**: 1st place as Technical Lead (see Makercie).
- **International Collegiate Programming Contest (2022, 2023)**: 28th in Benelux, 58th in Northwestern Europe.
- **Beta Business Days Startup Competition (2024)**: 1st place, following a rapid development of a viable business strategy.

Skills

Programming: Python, C, Java, SQL, Bash, Linux
Machine Learning: PyTorch, Keras, scikit-learn, Hugging Face Transformers, LoRA fine-tuning, LLMs, BERT, Diffusion Models, Reinforcement Learning, NLP, Explainable AI
Computer Vision & Robotics: SLAM, 3D Gaussian Splatting, Structure from Motion, OpenCV, ROS, Gazebo, Isaac Sim, Isaac Lab, Isaac Gym, Sim-to-Real, World Models
Data & Infrastructure: NumPy, Pandas, Matplotlib, Parquet, Docker, FastAPI, SLURM/HPC, WandB, MongoDB, React, Node.js
Engineering Practice: Git, GitLab CI/CD, Unit Testing, Agile/Scrum

Languages

Polish: Native
English: Full Professional Proficiency

Volunteering

Volunteer Firefighter **Poznan** *2017 – 2021*

Emergency Response Officer **Cover Student Association** *2022 – 2024*